

01

Code specifications

Cuts applied in Matteo's original code (per channel): MuTau

mutau				
	ttjets	dy	qcd	data
FASE0	1 tau. 1 muon	1 tau. 1 muon	1 tau. 1 muon	1 tau. 1 muon
Doube_medium				
Iso_Mu24	✓	✓	✓	✓
Ele32_WpTight				
Electron_				
lumi filtering			✓	✓
FASE1				
tau id1	>63	>63	>63	>63
tau id2	>7	>7	>7	>7
tau id3	>1	>1	>1	>1
tau pt	>100	>100	>100	>100
mu id	>=3	>=3	>=3	>=3
mu pt	>35	>35	>35	>35
ele id				
ele pt				
delta R	>0.4	>0.4	>0.4	>0.4
eta	<2.4	<2.4	<2.4	<2.4
charge signs	opposite	opposite	same	opposite
scaling factors	✓	✓		
valid jet and MET info	✓	✓		

Selection cuts

1 muon and 1 τ_h (ID+ISO)
 $p_T(\mu) > 35 \text{ GeV}, p_T(\tau_h) > 100 \text{ GeV}$
 Opposite sign (same-sign for QCD)
 At least two protons in PPS

Cuts explained in the analysis
 note
 proton requirements are applied
 after introducing pileup protons

02

Variable plots

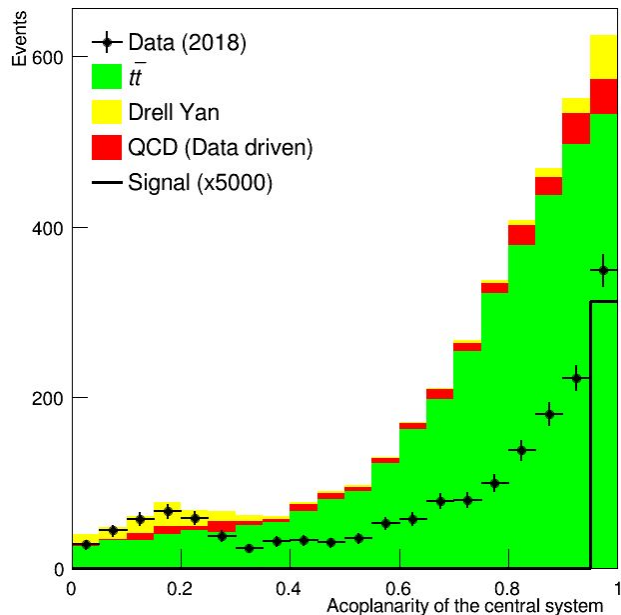
Input for the BDT

Input for the BDT; signal magnified for visualization

CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

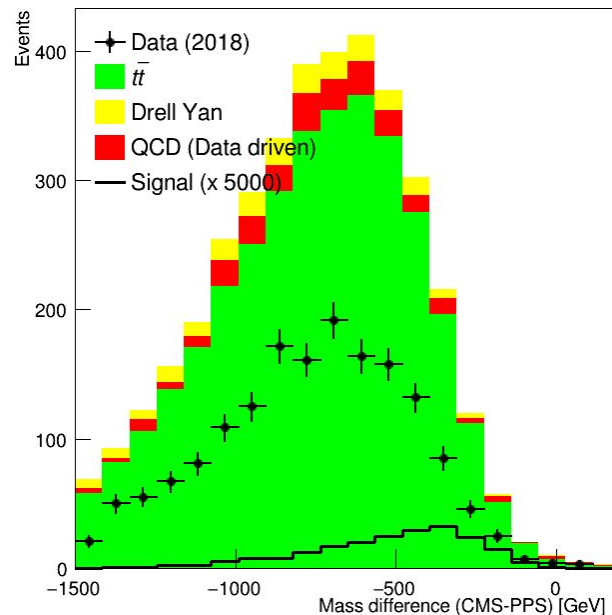
Preliminary



CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

Preliminary

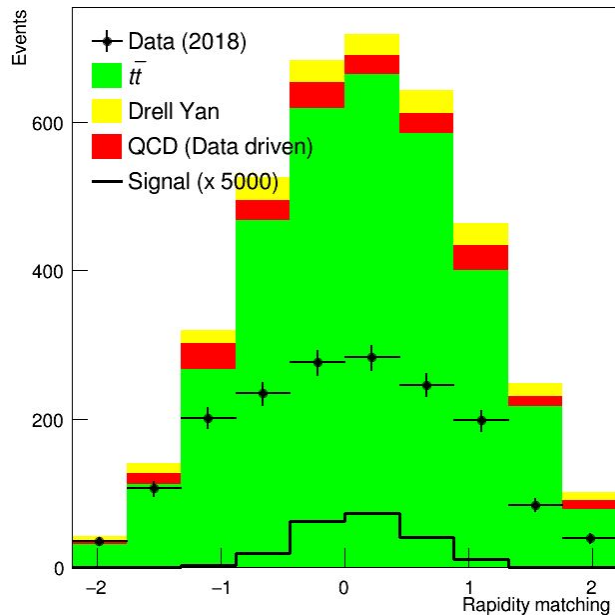


Input for the BDT; signal magnified for visualization

CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

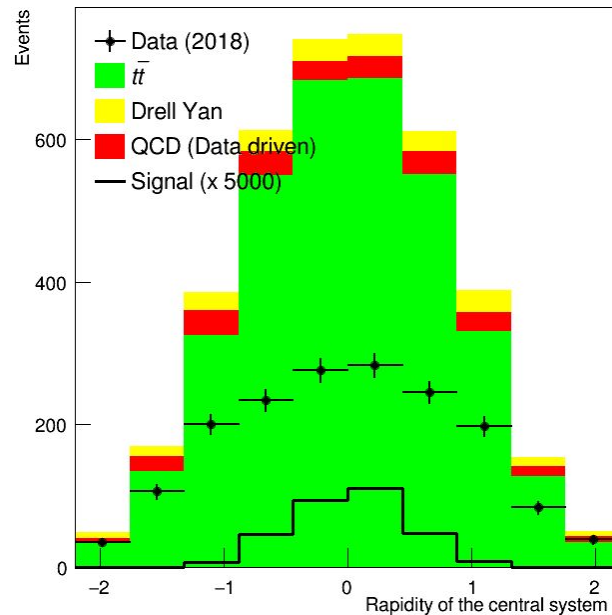
Preliminary



CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

Preliminary

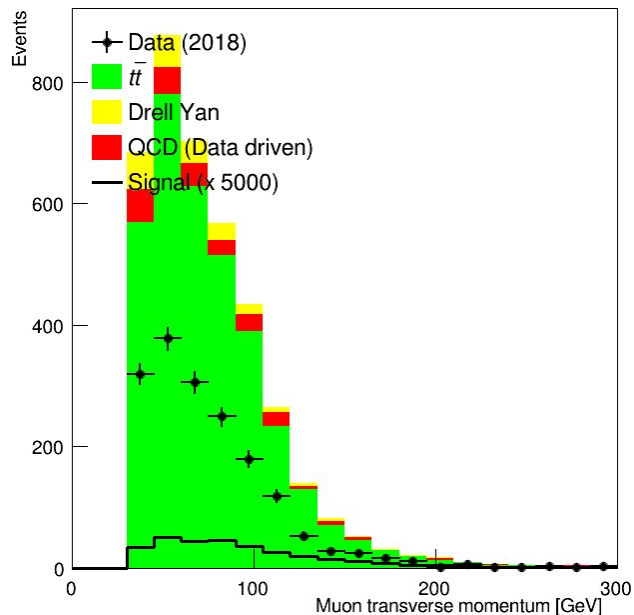


Input for the BDT; signal magnified for visualization

CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

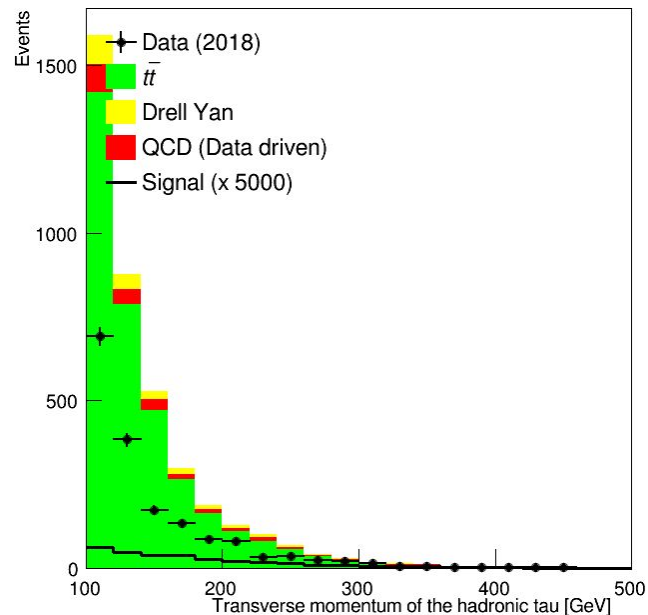
Preliminary



CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

Preliminary

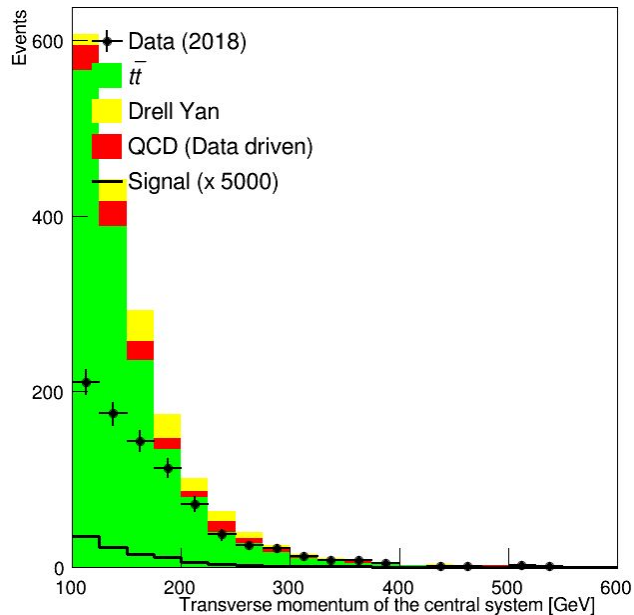


Input for the BDT; signal magnified for visualization

CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

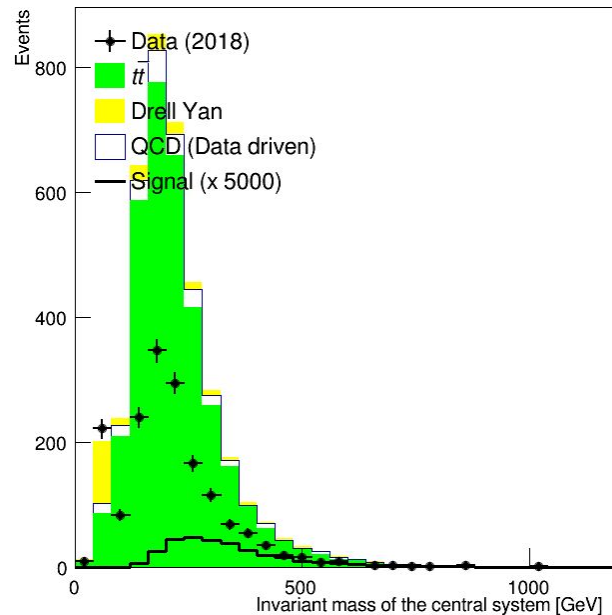
Preliminary



CMS-TOTEM

54.9 fb⁻¹ (13 TeV)

Preliminary

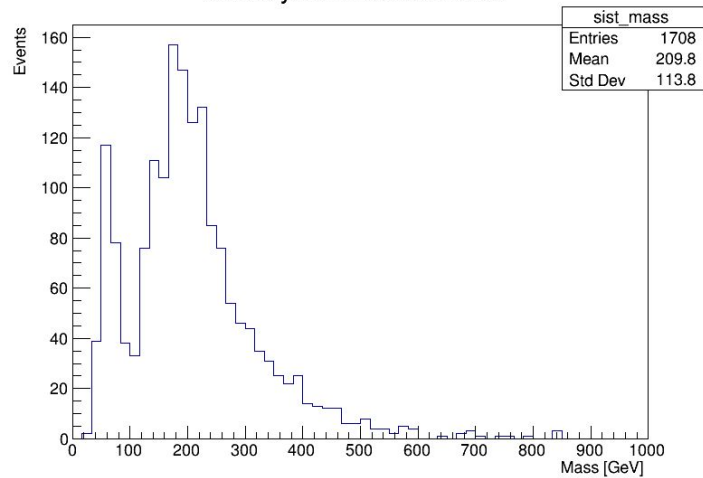


02

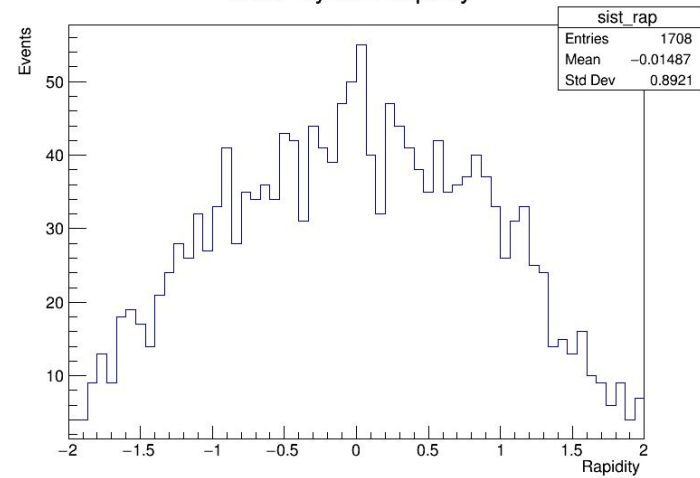
Variable plots

DATA

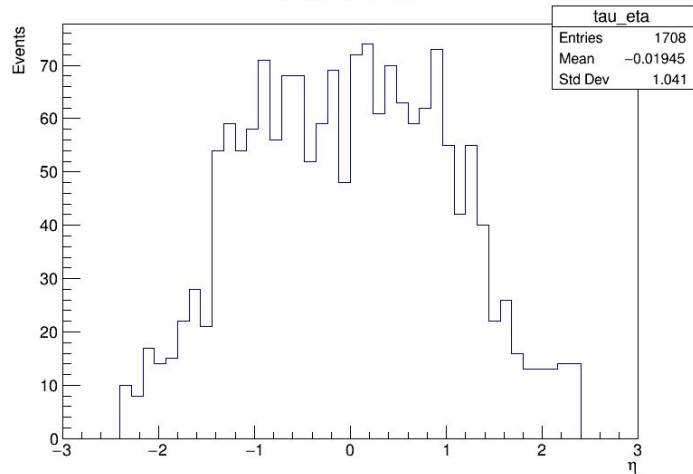
Data: System Invariant Mass



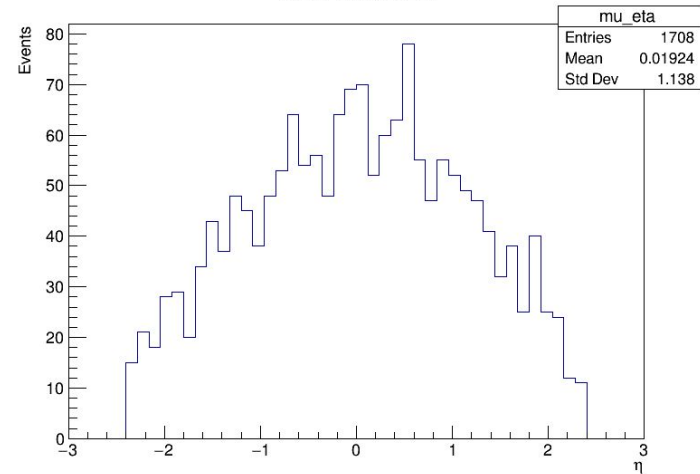
Data: System Rapidity



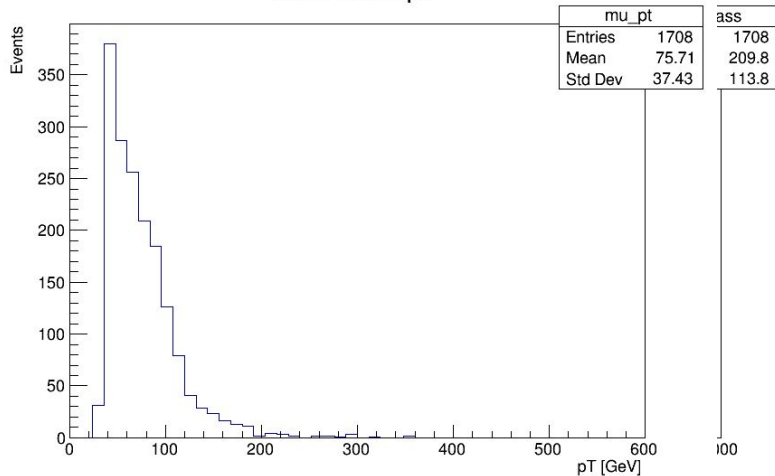
Data: Tau Eta



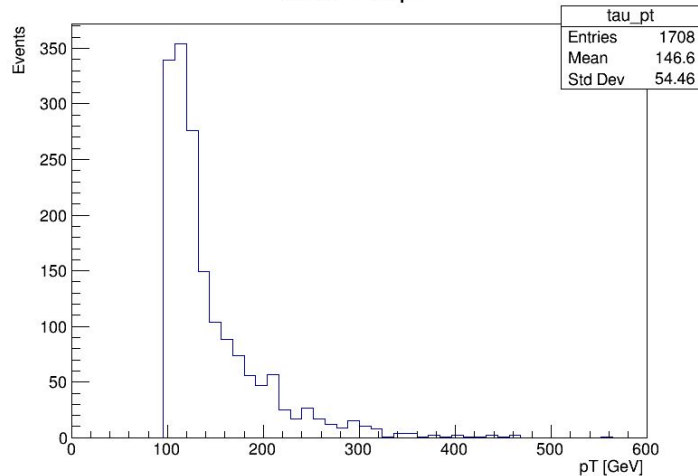
Data: Muon Eta



Data: Muon pT



Data: Tau pT



Sample details: Data with these cuts applied

data	
1 tau. 1 muon	
✓	
✓	
>63	
>7	
>1	
>100	
>=3	
>35	
>0.4	
<2.4	
opposite	

02

Variable plots

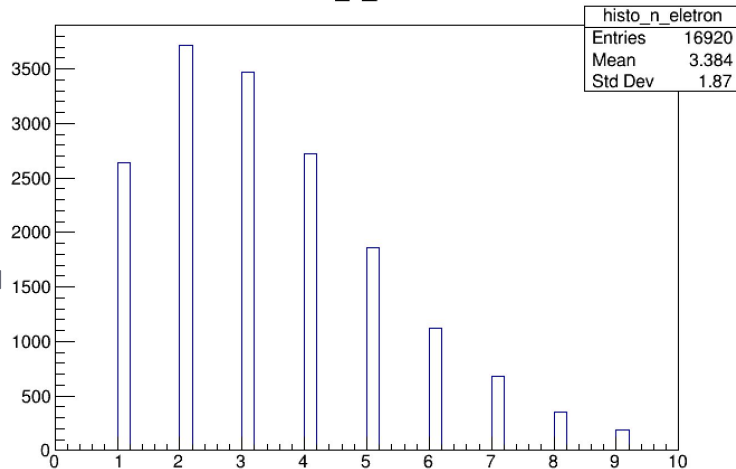
Signal

Signal plots
produced by
sinal.cpp code
mutau channel

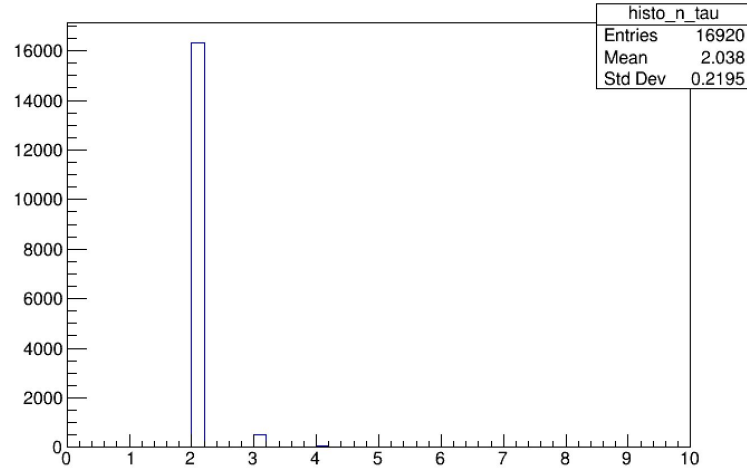
note: where is written e
or *electron* it should read
muon

- μ pT > 35
- tau pT > 100
- $\eta < 2.4$ for both
- opposite sign candidates
- at least one proton per pps arm

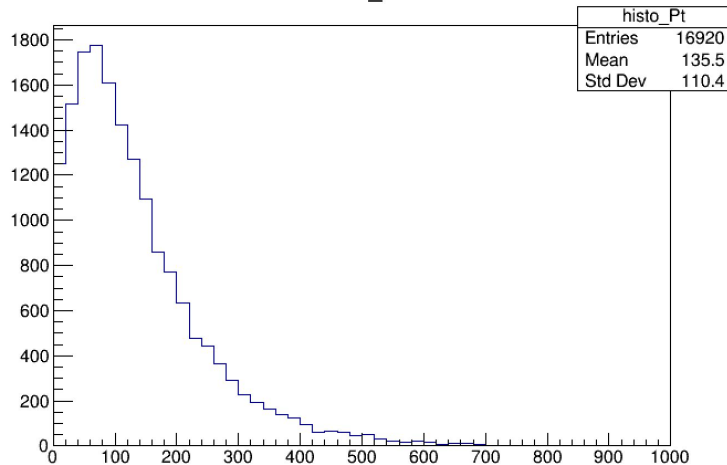
histo_n_eletron



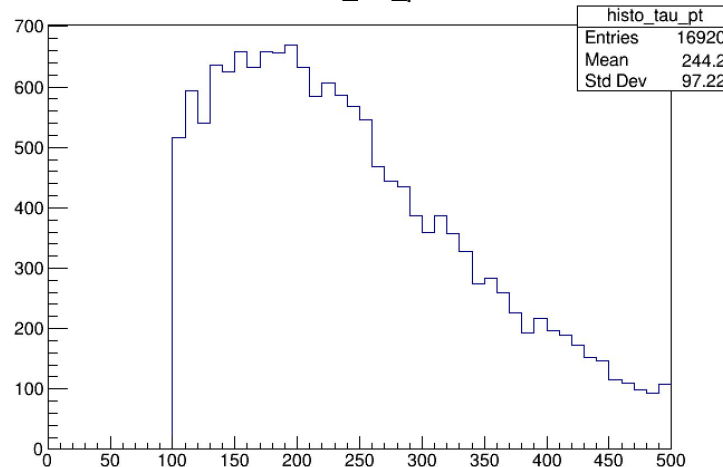
histo_n_tau



histo_Pt



histo_tau_pt

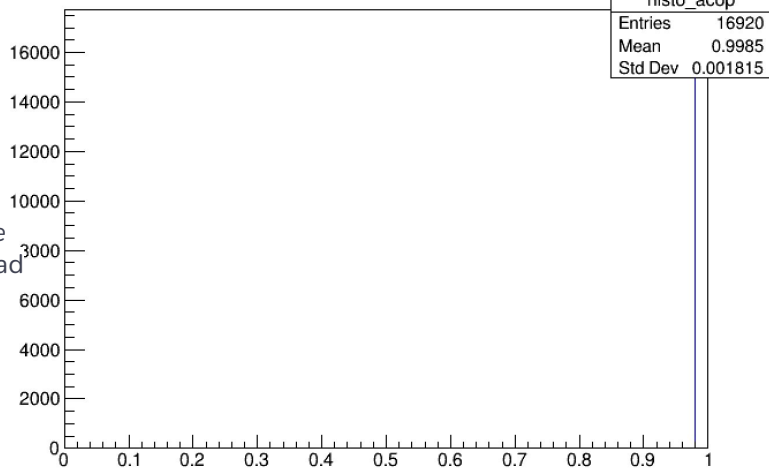


Signal plots
produced by
sinal.cpp code
mutau channel

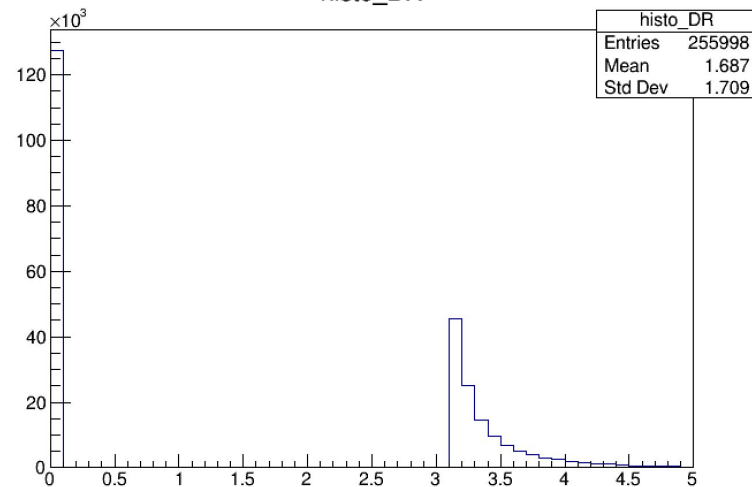
note: where is written e
or *electron* it should read
muon

- mu $p_T > 35$
- tau $p_T > 100$
- $|\eta| < 2.4$ for both
- opposite sign candidates
- at least one proton per pps arm

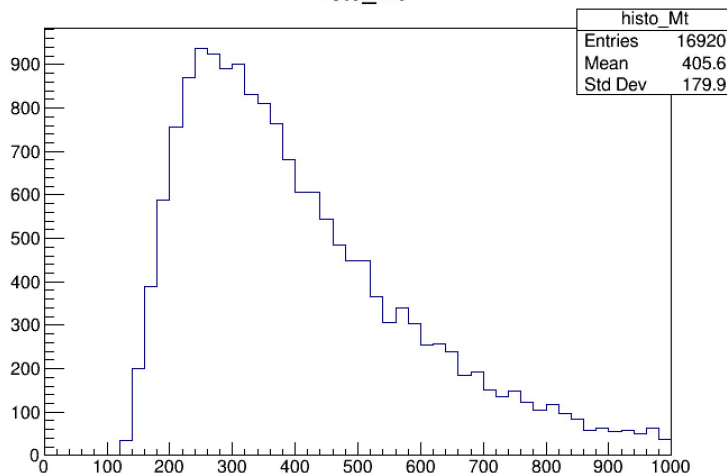
histo_acop



histo_DR



histo_Mt



histo_e_pt

